

## TUAPSE, Russian Federation

WMO: 370180

Lat: 44.100N

Lon: 39.067E

Elev: 62

StdP: 100.59

Time Zone: 3.00 (E03)

Period: 94-18

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -5.3       | -2.7 | -13.6                         | 1.2 | -3.6 | -10.4 | 1.6 | -1.4 | 11.6                  | 9.4  | 10.7 | 8.2  | 5.4                      | 40   | 0.399 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 8.0                    | 32.4            | 23.8 | 30.9 | 23.5 | 29.6 | 22.9 | 25.7                | 30.1 | 24.8 | 29.1 | 24.0 | 28.4 | 2.6                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 24.4                            | 19.5 | 29.2 | 23.5 | 18.4 | 28.3 | 22.5 | 17.4 | 27.4 | 80.0          | 30.0 | 76.1 | 29.1 | 72.6 | 28.4 | 28.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| 10.3              | 8.9  | 7.4 | DB                         | -7.6 | 34.6               | 3.0 | 1.8                                                | -9.7  | 35.9       | -11.4 | 37.0       | -13.1 | 38.0       | -15.3 | 39.4 | (4) |
|                   |      |     | WB                         | -9.2 | 26.8               | 2.8 | 1.3                                                | -11.2 | 27.7       | -12.9 | 28.5       | -14.4 | 29.2       | -16.5 | 30.1 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 14.8  | 5.8   | 6.3   | 8.5   | 12.6  | 17.5  | 21.3  | 24.6  | 25.3  | 21.3  | 15.9  | 11.1  | 7.2  | (6)  |
| (7)  |                                                                                  | DBStd       | 7.74  | 4.17  | 4.14  | 3.51  | 3.47  | 3.42  | 2.68  | 2.52  | 2.35  | 3.05  | 3.54  | 3.96  | 4.45 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 480   | 137   | 113   | 72    | 13    | 0     | 0     | 0     | 0     | 3     | 34    | 108   |      | (8)  |
| (9)  |                                                                                  | HDD18.3     | 1933  | 389   | 338   | 305   | 175   | 57    | 6     | 0     | 9     | 90    | 217   | 347   |      | (9)  |
| (10) |                                                                                  | CDD10.0     | 2246  | 6     | 9     | 25    | 91    | 232   | 341   | 453   | 476   | 340   | 187   | 67    | 20   | (10) |
| (11) |                                                                                  | CDD18.3     | 656   | 0     | 0     | 0     | 3     | 31    | 97    | 194   | 217   | 99    | 14    | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 4772  | 0     | 0     | 0     | 11    | 159   | 477   | 1613  | 1933  | 536   | 43    | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 1364  | 0     | 0     | 0     | 1     | 32    | 84    | 489   | 642   | 113   | 3     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.1   | 3.8   | 3.8   | 3.5   | 2.9   | 2.3   | 2.4   | 2.4   | 2.5   | 2.9   | 3.1   | 3.6   | 3.9  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 1528  | 173   | 129   | 119   | 93    | 103   | 106   | 96    | 95    | 125   | 154   | 158   | 177  | (15) |
| (16) |                                                                                  | PrecMax     | 1977  | 265   | 268   | 269   | 186   | 276   | 246   | 336   | 268   | 379   | 389   | 329   | 364  | (16) |
| (17) |                                                                                  | PrecMin     | 1042  | 50    | 39    | 55    | 36    | 16    | 24    | 4     | 5     | 18    | 27    | 21    | 33   | (17) |
| (18) |                                                                                  | PrecStd     | 208   | 56    | 62    | 47    | 41    | 66    | 52    | 80    | 76    | 76    | 84    | 84    | 74   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 14.9  | 16.1  | 19.8  | 24.9  | 30.0  | 30.8  | 34.1  | 34.7  | 31.9  | 26.7  | 21.3  | 18.2 | (19) |
| (20) |                                                                                  | 0.4%        | MCWB  | 8.8   | 8.8   | 11.2  | 15.0  | 19.3  | 21.8  | 24.9  | 24.8  | 23.4  | 19.3  | 13.5  | 12.2 | (20) |
| (21) |                                                                                  | 2%          | DB    | 13.3  | 14.5  | 17.3  | 22.0  | 26.9  | 28.8  | 32.2  | 32.7  | 29.5  | 24.4  | 19.5  | 16.1 | (21) |
| (22) |                                                                                  | 2%          | MCWB  | 8.4   | 8.5   | 9.9   | 13.6  | 18.9  | 22.1  | 24.0  | 24.2  | 22.0  | 18.1  | 13.2  | 10.5 | (22) |
| (23) |                                                                                  | 5%          | DB    | 12.3  | 13.2  | 15.2  | 19.9  | 24.9  | 27.3  | 30.8  | 31.2  | 27.8  | 22.8  | 18.1  | 14.6 | (23) |
| (24) |                                                                                  | 5%          | MCWB  | 8.1   | 8.3   | 9.4   | 13.0  | 18.2  | 21.7  | 23.7  | 23.8  | 21.1  | 17.2  | 13.0  | 9.7  | (24) |
| (25) |                                                                                  | 10%         | DB    | 11.2  | 11.8  | 13.6  | 18.1  | 23.0  | 26.0  | 29.4  | 29.9  | 26.3  | 21.2  | 16.8  | 13.2 | (25) |
| (26) |                                                                                  | 10%         | MCWB  | 7.7   | 7.7   | 8.9   | 12.3  | 17.7  | 21.1  | 23.1  | 23.2  | 20.5  | 16.9  | 12.6  | 9.3  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 10.5  | 10.4  | 12.0  | 16.1  | 21.3  | 24.4  | 27.0  | 27.0  | 25.4  | 20.2  | 16.0  | 13.4 | (27) |
| (28) |                                                                                  | 0.4%        | MCDWB | 12.8  | 13.0  | 17.1  | 21.3  | 26.0  | 28.2  | 31.7  | 32.7  | 29.2  | 24.6  | 18.4  | 17.3 | (28) |
| (29) |                                                                                  | 2%          | WB    | 9.6   | 9.6   | 10.9  | 14.9  | 20.0  | 23.3  | 25.8  | 25.9  | 23.2  | 19.2  | 15.0  | 11.5 | (29) |
| (30) |                                                                                  | 2%          | MCDWB | 11.9  | 12.6  | 15.4  | 19.9  | 24.5  | 27.4  | 30.0  | 30.6  | 27.9  | 23.1  | 17.9  | 14.4 | (30) |
| (31) |                                                                                  | 5%          | WB    | 8.9   | 8.9   | 10.1  | 13.9  | 19.1  | 22.4  | 24.9  | 25.0  | 22.0  | 18.1  | 14.0  | 10.6 | (31) |
| (32) |                                                                                  | 5%          | MCDWB | 11.3  | 12.0  | 13.6  | 18.5  | 23.6  | 26.4  | 29.2  | 29.3  | 26.6  | 21.6  | 17.0  | 13.5 | (32) |
| (33) |                                                                                  | 10%         | WB    | 8.1   | 8.2   | 9.4   | 13.0  | 18.2  | 21.6  | 24.1  | 24.2  | 21.0  | 17.2  | 13.0  | 9.7  | (33) |
| (34) |                                                                                  | 10%         | MCDWB | 10.8  | 11.3  | 12.6  | 17.1  | 22.4  | 25.3  | 28.3  | 28.7  | 25.4  | 20.5  | 16.1  | 12.5 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | MDBR        | 5.2   | 5.5   | 5.6   | 6.5   | 7.1   | 6.8   | 7.4   | 8.0   | 7.9   | 7.2   | 6.7   | 5.5   | (35) |      |
| (36) |                                                                                  | 5% DB       | MCDBR | 6.2   | 6.9   | 8.6   | 9.8   | 10.2  | 8.4   | 8.9   | 9.5   | 9.8   | 9.5   | 8.9   | 6.8  | (36) |
| (37) |                                                                                  | 5% DB       | MCWBR | 4.1   | 3.9   | 4.3   | 4.7   | 4.9   | 4.8   | 5.2   | 5.5   | 5.8   | 5.5   | 5.9   | 4.7  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 5.2   | 5.4   | 6.9   | 8.0   | 8.5   | 7.4   | 7.8   | 8.2   | 8.9   | 8.1   | 7.5   | 5.9  | (38) |
| (39) |                                                                                  | 5% WB       | MCWBR | 3.8   | 3.7   | 4.0   | 4.5   | 4.7   | 4.8   | 5.0   | 5.3   | 5.7   | 5.4   | 5.6   | 4.6  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.350 | 0.378 | 0.424 | 0.462 | 0.430 | 0.429 | 0.446 | 0.447 | 0.407 | 0.389 | 0.358 | 0.345 | (40) |      |
| (41) |                                                                                  | taud        | 2.318 | 2.236 | 2.137 | 2.081 | 2.271 | 2.324 | 2.290 | 2.283 | 2.370 | 2.369 | 2.385 | 2.352 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 751   | 793   | 802   | 803   | 844   | 845   | 825   | 810   | 816   | 776   | 740   | 719   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 83    | 107   | 135   | 154   | 131   | 125   | 128   | 125   | 106   | 94    | 78    | 74    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.31  | 2.16  | 3.10  | 4.68  | 5.82  | 6.49  | 6.62  | 5.84  | 4.38  | 2.84  | 1.74  | 1.17  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.18  | 0.26  | 0.26  | 0.52  | 0.64  | 0.43  | 0.33  | 0.36  | 0.43  | 0.26  | 0.20  | 0.15  | (45) |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | +0.69 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (5 neighbors) | +0.59 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +134    | +70     |

Nomenclature: See separate page